Beyond Small Corrections: Dynamical Unitarity and the Nature of the Black Hole Information Paradox

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(Dated: July 20, 2026)

Abstract

We argue that critiques of Mathur's small corrections theorem based on the Eigenstate Thermalization Hypothesis (ETH) and quantum gravitational state corrections commit a category error, conflating kinematic corrections to the initial state with dynamic corrections to the evolution operator. The true burden of the information paradox lies in the latter. We show that granting ETH in full does not evade this burden. The small corrections theorem itself already assumes that the effective, macroscopic side of the description evolves unitarily on a fixed effective Hilbert space at each time; this is simply what it means to have an EFT description in the first place. Under this same, reasonable assumption, even if the true late-time microstate is kinematically far from any naive semiclassical description, wherever the effective evolution remains locally ordinary ('no drama'), the small corrections theorem's entropy-growth logic still applies, regardless of how the underlying microstate is dressed. We complement this dynamical response with a *Page Curve Norm Requirement*: producing the Page curve turnaround forces the operator norm of the interaction perturbation to be macroscopically large on the emission timescale, independent of which state realizes it. Together, these results imply that all prominent proposed resolutions, including black hole complementarity, ER=EPR, and replica wormholes, implicitly concede the theorem by requiring macroscopic dynamical departures from local effective field theory.

Keywords: small corrections theorem, no-drama condition, dynamical unitarity, black hole information problem, kinematic vs. dynamic corrections, eigenstate thermalization hypothesis

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I. INTRODUCTION

The black hole information paradox, since its inception by Stephen Hawking [1], has remained a central driver of high-energy and black hole physics. One of the sharpest modern formulations of this paradox is articulated by Samir D. Mathur's small corrections theorem [2]. Mathur mathematically demonstrated that if one assumes the emission of Hawking radiation follows semiclassical Effective Field Theory (EFT) to leading order, small, local ϵ-corrections to the creation of entangled pairs cannot accumulate to restore the purity of the final state.

Despite its clarity, the theorem is often dismissed or conceptually bypassed by researchers appealing to the full apparatus of quantum gravity (QG). A prevalent counter-argument asserts that QG naturally provides large corrections to the initial EFT state. Concurrently, the Eigenstate Thermalization Hypothesis (ETH) [3] is frequently invoked to suggest that while the true QG microstate differs wildly from the EFT state, they yield identical expectation values for "simple" observables. Therefore, the critique goes, any purely EFT-based result, including Mathur's theorem, is invalidated by the presence of large quantum gravitational state corrections. A version of this line of argument, along with a broader survey of related critiques, is developed by Raju [4], who argues that only exponentially small correlations in the outgoing radiation are needed to resolve Hawking's original paradox.

This paper pursues two related goals. The first is to clarify why such dismissals commit a category error, conflating *kinematic* (or static) corrections with *dynamic* corrections. The small corrections theorem does not fail simply because the initial state requires significant alteration. Instead, it asks whether the step-by-step *evolution* of the state requires macroscopic correction. The second goal is to show that this distinction, once made precise, already answers the ETH objection on its own terms, under the same reasonable assumption the small corrections theorem itself makes: that the effective, macroscopic side of the description evolves unitarily on a fixed effective Hilbert space at each time. Granting ETH does not require abandoning this assumption; it only concedes that the true microstate is kinematically complex. As long as an old black hole's horizon remains locally ordinary ('no drama') at a given moment, some unitary EFT, however corrected or dressed relative to the leading-order Hawking picture, still governs that moment's dynamics on the same effective Hilbert space, and the small corrections theorem's entropy-growth logic applies to it directly.

This argument is state-independent in the relevant sense: it does not depend on which kinematic corrections dress the true microstate, only on whether, under the assumption above, the effective side remains dynamically unitary and no-drama at each time.

Framed physically, the entire dispute can be stated as a single question: does the macroscopic, effective description of the black hole remain unitary and linear *dynamically*, at every moment of the evaporation process, or only in the trivial sense that some global unitary map connects the initial and final states? An EFT-respecting picture does not require that the semiclassical approximation stay exact forever; local effective field theory itself anticipates corrections at late times, once entanglement structures become extreme. What an EFT-respecting picture assumes, in keeping with the premise the small corrections theorem itself relies on, is that at each time t there is a well-defined, approximately unitary generator describing the system, together with correction terms whose overlap with the semiclassical branch stays controlled rather than appearing from nowhere. Section III B makes this precise: the condition that the growing record of early radiation carries negligible weight within the semiclassical branch must itself evolve smoothly, rather than jump discontinuously from small to large exactly at the Page time.

Together, the small corrections theorem, the no-drama response to ETH, and the Page Curve Norm Requirement demonstrate that all physical resolutions to the information paradox, including Black Hole Complementarity, the ER=EPR conjecture, and replica wormholes, implicitly concede these results. They do so by requiring macroscopic dynamical alterations to local effective field theory, or by abandoning the local tensor product structure of spacetime entirely.

We set aside a detailed evaluation of AdS/CFT-based dismissals of the paradox, since Mathur himself has already addressed why such dismissals fail, and for a reason that has nothing to do with contesting the unitarity of the boundary theory. In a pedagogical dialogue between a skeptical student and a "Hawking believer" [2], Mathur shows that appealing to the unitarity of the CFT to dismiss the paradox is exactly as circular as appealing to the unitarity of laboratory quantum mechanics: neither move explains *how* the local, near-horizon dynamics manages to stay unitary, which is the entire content of the paradox. Knowing that some global map is unitary does not by itself specify the local mechanism, so the burden the small corrections theorem identifies survives the invocation of AdS/CFT untouched.

II. THE SMALL CORRECTIONS THEOREM AND THE ETH OBJECTION

In the standard EFT description, the vacuum near the horizon produces a pair of quanta: an exterior mode b_1 and an interior mode c_1. To leading order, they are maximally entangled:

$$|\Psi\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle_{b_1} |0\rangle_{c_1} + |1\rangle_{b_1} |1\rangle_{c_1} \right). \tag{1}$$

The entanglement entropy of the early radiation $S(B_N)$ grows by $\ln 2$ with each emission. Mathur's theorem posits that if the true state includes a small correction, $|\Psi\rangle_{\text{exact}} = |\Psi\rangle_{\text{EFT}} + |\delta\Psi\rangle$ where $||\delta\Psi|| < \epsilon$, the change in entanglement entropy remains bounded:

$$S(B_{N+1}) > S(B_N) + \ln 2 - 2\epsilon. \tag{2}$$

Thus, small corrections cannot induce the Page curve turnaround required for unitary evaporation.

A common philosophical and physical objection leverages ETH and the vastness of the QG Hilbert space. Because the Bekenstein-Hawking entropy dictates an exponentially large number of microstates ($e^{S_{BH}}$), it is argued that the semiclassical EFT vacuum is merely a coarse-grained thermodynamic approximation. ETH implies that many dramatically different pure states in the underlying QG theory will reproduce the same simple observable expectation values as the EFT state. Because the true state involves $\mathcal{O}(1)$ (large) corrections to the EFT description, critics claim the ϵ-bound of the small corrections theorem is fundamentally inapplicable.

Section III A answers this objection directly: as long as the horizon remains locally ordinary ('no drama') at a given time, some unitary EFT on the same effective Hilbert space governs that time's dynamics regardless of how exotic the underlying microstate is, and the entropy continues to climb under it. Section IV then shows, independently and without any appeal to ETH, that the Page curve's $\mathcal{O}(1)$ entropy change forces a macroscopically large operator norm for the perturbation that produces it. Both conclusions are immune to ETH-style arguments about microstate complexity.

III. KINEMATIC VS. DYNAMIC CORRECTIONS

The aforementioned critique misidentifies the operational core of Mathur's argument. To see why, we must establish a rigorous distinction between correcting the state (*kinematics*) and correcting the evolution operator (*dynamics*).

We can readily concede that the initial state of the black hole, $|\Psi_0\rangle$, may be subject to arbitrarily large corrections due to QG microstate structure. We can replace the naive EFT vacuum with any highly complex QG state. However, the critical question is what happens to this state under time evolution.

Let U_{exact} be the evolution operator of the exact quantum gravity theory, and let U_{EFT} be the evolution operator of the effective field theory that Hawking's original derivation assumes governs the horizon. Over most of the relevant subspace, U_{EFT} is an excellent representative of U_{exact}; the two agree closely, except possibly on some subset of states, where they can differ substantially. Over a single emission step of duration Δt we can write:

$$U_{\text{exact}} \approx U_{\text{EFT}} + \delta U. \tag{3}$$

Because U_{exact} is fixed by the exact theory while the black hole's state changes as it evaporates, the particular EFT operator that best represents U_{exact} can itself change from one emission step to the next. It is therefore more accurate to let the effective generator carry a time label, $U_{t,\text{EFT}}(\Delta t)$, and to allow it to differ from its counterpart at an earlier time by as much as $\mathcal{O}(1)$, rather than to hold it fixed to its leading-order Hawking form throughout the entire evaporation. What the small corrections theorem needs is not that $U_{t,\text{EFT}}(\Delta t)$ stay literally fixed, but that at each time t it remain some local, unitary operator on a fixed effective Hilbert space $\mathcal{H}_{\text{EFT}}$. A large correction, $\|\delta U\| \sim 1$, does not by itself violate this: it can simply mean that the locally relevant generator has been updated from $U_{t_1,\text{EFT}}(\Delta t)$ to a substantially different $U_{t_2,\text{EFT}}(\Delta t)$, while local unitarity and the effective Hilbert space are both preserved. What $\delta U \sim 1$ threatens is not unitarity itself, but the assumption that a single, fixed EFT generator continues to represent U_{exact} throughout the evaporation. Section III A addresses what is required to keep $U_{t,\text{EFT}}(\Delta t)$ well-defined, local, and unitary once $\delta U \sim 1$.

The ETH objection establishes that $|\Psi_0\rangle_{\text{exact}} \neq |\Psi_0\rangle_{\text{EFT}}$. But Mathur's theorem does not require the initial state to be purely EFT. It requires that the *action* of generating the next Hawking quantum depends on the dynamics of the horizon.

If the dynamic correction is small at a given time, that is, if $U_{t,\text{EFT}}(\Delta t)$ only deviates from U_{exact} by an operator norm $\|\delta U\| \sim \epsilon$ over that emission step, then the bound on the entanglement entropy holds at that step. Even if we completely replace the initial static state to align with the "right" QG microstate, a local evolution that is approximately semiclassical ($\delta U \ll 1$) will continue to generate entanglement monotonically.

Therefore, invoking ETH or QG Hilbert space complexity to argue that the EFT *state* is vastly incorrect is a non-sequitur regarding the information paradox. To save unitarity, the theorem dictates that one must drastically alter the *dynamics*, in the sense of forcing $U_{t,\mathrm{EFT}}(\Delta t)$ to stop being a local, unitary operator on the fixed effective Hilbert space $\mathcal{H}_{\mathrm{EFT}}$, and not merely in the weaker sense of updating which local EFT generator happens to be relevant at a given time. The power of the small corrections theorem lies precisely in forcing this dichotomy: static corrections are insufficient, and an evolving but still local and unitary $U_{t,\mathrm{EFT}}(\Delta t)$ is insufficient too; only a genuine breakdown of local unitarity or of the effective Hilbert space itself can save unitarity.

A. The No-Drama Response to the ETH Objection

The kinematic/dynamic distinction developed above already answers the ETH objection in full, without requiring any additional operator-norm bound on top of it, provided we adopt the same reasonable assumption the small corrections theorem itself makes: that at each time t the effective, macroscopic side of the description is governed by some $U_{t,\mathrm{EFT}}(\Delta t)$ that is local and unitary on a fixed effective Hilbert space. This is not an extra postulate introduced to rescue the argument; it is simply the working sense in which an EFT description is meaningful at all, and it is what the small corrections theorem already presupposes when it speaks of an EFT description in the first place. Grant the ETH picture in its strongest form: the true quantum gravitational microstate of an old black hole is astronomically different from any single semiclassical description, and this difference is invisible to low-energy observables. Neither claim requires abandoning the assumption above, and the following two points follow from it.

First, an old black hole's exterior is still expected to look locally like an ordinary horizon to an observer falling in. This is the equivalence principle applied to the horizon, sometimes called the no-drama condition. A horizon that has just formed on a small, stable black hole is a mundane, low-curvature region, and general relativity together with local effective field theory describes it accurately. Think of the near-horizon region of a young, freshly formed black hole. If an old, evaporating black hole's horizon remains locally similar in curvature scale and local vacuum structure, the same no-drama expectation applies there: an infalling observer should encounter ordinary physics, governed by some local, unitary effective field theory, whatever exotic microstate that local physics happens to sit on top of.

Second, and this is the part ETH does not touch, if we hold to the assumption stated above, then whatever $U_{t,\mathrm{EFT}}(\Delta t)$ is at a given late time, it acts on the same effective-degree Hilbert space that the original Hawking derivation used, and it is unitary on that space simply because the assumption above already requires it. That the true microstate is complicated does not by itself license the further claim that this assumption has broken down, that the effective, macroscopic evolution rule governing the microstate's low-energy degrees of freedom has stopped being an ordinary local unitary EFT. Nothing forces $U_{t,\mathrm{EFT}}(\Delta t)$ to be identical to the leading-order EFT generator that Hawking used; it is free to be corrected, updated, or replaced by a heavily dressed effective description as the black hole ages, possibly by an $\mathcal{O}(1)$ amount from one emission step to the next. What it cannot do, so long as no-drama holds and the assumption above is maintained, is fail to be *some* unitary EFT generator on the same effective Hilbert space at each time step.

Once this shared assumption is granted, the small corrections theorem's logic reattaches itself immediately. At any time t at which the horizon is no-drama, the locally relevant $U_{t,\mathrm{EFT}}(\Delta t)$ generates entanglement between the black hole and its radiation in exactly the way Hawking's original derivation describes, regardless of which corrected or dressed generator happens to be in force, because the theorem's entropy-growth bound depends only on the smallness of the dynamic correction δU over one emission step, not on the specific content of $U_{t,\mathrm{EFT}}(\Delta t)$ at that step. ETH-style kinematic complexity in the state changes which EFT is locally relevant; it does not, by itself, stop that EFT from being unitary and local, and it is exactly this residual unitarity and locality, present at every no-drama time, that keeps the entropy climbing. To reverse the climb, the horizon must eventually stop being no-drama: a genuine dynamical departure must occur, whether a violation of linearity, a macroscopic phase transition in the state's amplitudes, or a breakdown of the effective Hilbert space factorization itself, as catalogued in Section III B. ETH supplies none of these on its own; it only supplies kinematic distance between microstates, and kinematic distance alone cannot force a dynamical breakdown of no-drama unitarity.

This is also the right way to state what ETH does and does not achieve in terms of the overlap language developed in Section III B. Early corrections to the state, of exactly the kind ETH supplies, do not by themselves force the overlap $\langle\psi_{\text{EFT-like}}(t)|\delta\Psi_{\text{early}}(t)\rangle$ to grow: they can leave the effective side no-drama and unitary, generating entropy exactly as before, while the true global state accumulates whatever kinematic complexity ETH predicts. The overlap can only be pushed toward $\mathcal{O}(1)$ by a dynamical event of the kind above. Early kinematic corrections alone cannot drive the large late-time correction that the Page curve demands.

B. Late-Time Dynamic Divergence and the Linearity Constraint

A more subtle attempt to evade the small corrections theorem involves positing a time-dependent divergence between the generator $U_{t,\text{EFT}}(\Delta t)$ that actually governs the state within the effective Hilbert space $\mathcal{H}_{\text{EFT}}$ and the leading-order semiclassical generator U_{EFT} that Hawking's original derivation assumed. One might hypothesize that for an early black hole state, $U_{t,\text{EFT}}(\Delta t)$ closely tracks the leading-order prediction: $U_{t,\text{EFT}}(\Delta t)|\Psi_{\text{early BH}}\rangle \approx U_{\text{EFT}}|\Psi_{\text{early BH}}\rangle$. However, as the black hole ages, specifically after the Page time, $U_{t,\text{EFT}}(\Delta t)$ diverges significantly from that leading-order form: $U_{t,\text{EFT}}(\Delta t)|\Psi_{\text{late BH}}\rangle \neq U_{\text{EFT}}|\Psi_{\text{late BH}}\rangle$.

This framework is highly appealing because it preserves the success of Hawking's original derivation for young black holes while attempting to salvage unitarity at late times. However, assessing the plausibility of this evasion requires confronting the linearity of quantum mechanics and the nature of macroscopic superpositions.

A late-time black hole state, viewed within the effective Hilbert space $\mathcal{H}_{\text{EFT}}$, is generically a massive superposition of microstates. Suppose we express this late state in a basis that includes branches resembling smooth, semiclassical EFT geometries alongside exotic, heavily quantum-gravitational (QG) microstates, all still elements of $\mathcal{H}_{\text{EFT}}$:

$$|\Psi_{\text{late BH}}\rangle = \alpha|\psi_{\text{EFT-like}}\rangle + \beta|\psi_{\text{QG-like}}\rangle. \tag{4}$$

By the principle of quantum linearity, the generator $U_{t,\text{EFT}}(\Delta t)$ must distribute over this superposition:

$$U_{t,\text{EFT}}(\Delta t)|\Psi_{\text{late BH}}\rangle = \alpha\, U_{t,\text{EFT}}(\Delta t)|\psi_{\text{EFT-like}}\rangle + \beta\, U_{t,\text{EFT}}(\Delta t)|\psi_{\text{QG-like}}\rangle. \tag{5}$$

To preserve the correspondence principle, the requirement that standard physics holds in regimes where semiclassical conditions are met, we must demand that $U_{t,\text{EFT}}(\Delta t)$

acts on the "normal" EFT branch exactly as the leading-order U_{EFT} would. That is, $U_{t,\text{EFT}}(\Delta t)|\psi_{\text{EFT-like}}\rangle \approx U_{\text{EFT}}|\psi_{\text{EFT-like}}\rangle$. This branch will therefore continue to strictly generate $\ln 2$ of entanglement entropy per emission step.

To achieve a unitary Page curve turnaround, the burden falls entirely on the β branches. These non-EFT outcomes must evolve in such a radically different way that their quantum interference perfectly cancels out the monotonically growing entanglement generated by the α branch.

Mathur's effective small corrections theorem [5] explicitly addresses this exact scenario, though the original theorem appended with the present section is largely sufficient for this purpose. The theorem demonstrates that if the amplitude of the semiclassical branch is dominant ($|\alpha|^2 \approx 1$), no amount of exotic evolution in the heavily suppressed β branches can mathematically overturn the entropy growth. The more fundamental reason behind this is unitarity itself: because $|\psi_{\text{EFT-like}}\rangle$ and $|\psi_{\text{QG-like}}\rangle$ are taken to be nearly orthogonal, $\langle\psi_{\text{EFT-like}}|\psi_{\text{QG-like}}\rangle \approx 0$, a unitary generator cannot transfer weight between them, so the branch amplitudes α and β stay essentially fixed regardless of how exotic the evolution within the β branch itself is. The strict subadditivity inequalities give an independent, quantitative bound on the influence of the β terms on the entropy, but it is this unitarity-plus-orthogonality argument that explains why α and β cannot simply be reassigned by the dynamics in the first place: doing so would itself require violating $\langle\psi_{\text{EFT-like}}|\psi_{\text{QG-like}}\rangle \approx 0$.

Therefore, for this time-dependent divergence to successfully purify the black hole, one of three highly radical physical realities seems to be required:

1. **Violation of Linearity:** The generator $U_{t,\text{EFT}}(\Delta t)$ acts non-linearly on $\mathcal{H}_{\text{EFT}}$, allowing the presence of the β branches to suppress the entanglement generation of the α branch without ordinary unitary interference.

2. **Macroscopic Dynamic Phase Transition:** The coefficient α must become small post-Page time. In other words, the late-time black hole state must possess small probability amplitude of being in a state that looks like a normal EFT geometry. By the point made above, this in turn requires that the near-orthogonality $\langle\psi_{\text{EFT-like}}|\psi_{\text{QG-like}}\rangle \approx 0$ itself be violated as the black hole evolves, since only such a violation would let unitary dynamics reduce $|\alpha|^2$.

3. **Breakdown of the Effective Hilbert Space Factorization:** The foundational

assumption that a fixed effective Hilbert space $\mathcal{H}_{\mathrm{EFT}}$ with a well-defined tensor product structure exists at all fails entirely at late times. There is no well-defined entanglement entropy of a black hole, and thus no well-defined Page curve, within any such fixed effective description.

The first option requires abandoning quantum mechanics on the effective side. The third option, understood as a breakdown of the fixed effective Hilbert space $\mathcal{H}_{\mathrm{EFT}}$ itself, is particularly relevant in the context of the AdS/CFT correspondence, where $\mathcal{H}_{\mathrm{EFT}}$ is realized concretely as the bulk Hilbert space. If this bulk factorization breaks down, there are currently two prevailing interpretations regarding its relationship to the unitary boundary theory:

Interpretation A (Matching Factorizations): The boundary and bulk Hilbert space factorizations are completely equivalent. If the boundary CFT does not permit a clean, local tensor factorization that isolates the 'interior' degrees of freedom from the early radiation, then the bulk quantum gravity theory simply cannot either. The apparent local factorization in the bulk is an illusion of the low-energy approximation, and U_{exact} inherently acts on a fundamentally non-factorizable space.

Interpretation B (Non-equivalent Factorizations via Subregion Holography): The boundary and bulk factorizations are fundamentally not equivalent. As demonstrated by Terashima's results on subregion holography [6], the localization of states differs drastically between the bulk and the boundary. A completely localized state within a boundary subregion does not map to a strictly localized state within the corresponding bulk entanglement wedge. Because the tensor product structure of the boundary conformal field theory does not isometrically translate to a localized tensor product structure in the bulk quantum gravity theory, the foundational assumption that the bulk Hilbert space cleanly factorizes (e.g., $\mathcal{H} = \mathcal{H}_{\mathrm{early}} \otimes \mathcal{H}_{\mathrm{out}} \otimes \mathcal{H}_{\mathrm{in}}$) breaks down completely. Consequently, an operator that appears perfectly local in the bulk (such as a local emission operator acting on the EFT branch) corresponds to a highly non-local, delocalized state on the boundary. This non-equivalence demonstrates that the strict spatial locality assumed by U_{EFT} is a flawed kinematic starting point for late-time black holes. This is not a completely negative result: it suggests that while the Page curve is ill-defined in the bulk, it can be imposed in the boundary. The $\mathcal{O}(1)$ turnaround in the Page curve now becomes more acceptable in the boundary, which contrasts with the circumstance in the bulk.

These three options are three ways of answering the same underlying question, namely whether the macroscopic, effective world remains unitary and linear *dynamically*, at each time step, rather than only in a coarse before-and-after sense. Nothing in the small corrections theorem or in this analysis requires that $U_{\rm EFT}$ remain exact forever. Effective field theory is itself expected to need corrections as the black hole approaches and passes the Page time; the target of the theorem is not that idea but the more specific claim that these corrections can remain dynamically small and local at every step while still summing to a macroscopic, unitarity-restoring effect. To make this concrete, write the correction terms that carry the entanglement record of previously emitted radiation as $|\delta\Psi_{\rm early}(t)\rangle$, and ask how their overlap with the semiclassical branch, $\langle\psi_{\text{EFT-like}}(t)|\delta\Psi_{\rm early}(t)\rangle$, behaves as a function of time. Early on this overlap is negligible; this is simply the small corrections theorem's premise restated. The open question posed by any late-time evasion is whether the overlap stays small and grows only gradually as the black hole ages, or whether it must instead jump from negligible to $\mathcal{O}(1)$ in the immediate vicinity of the Page time, precisely when the turnover is needed. A gradual, dynamically continuous growth of this overlap is exactly what a genuinely small, slowly accumulating operator norm $\|V(t)\|$ would produce, and Theorem 2 in Section IV shows that this is not sufficient to generate the $\mathcal{O}(1)$ Page curve turnaround. A sudden jump, on the other hand, is by definition a macroscopic dynamical discontinuity of exactly the kind the theorem forbids under approximately EFT dynamics. Either way, the requirement that the effective description remain unitary and linear at every time, not merely between the initial and final states, leaves no room for a resolution in which $U_{t,\rm EFT}(\Delta t)$ stays close to $U_{\rm EFT}$ at each step yet still purifies the radiation.

In all three scenarios, the attempt to smoothly evade the theorem fails. Even if we rely on the breakdown of bulk factorization, we are conceding that the local semiclassical operator $U_{\rm EFT}$ is structurally incapable of governing the system. One must still introduce a catastrophic, $\mathcal{O}(1)$ dynamic breakdown of the local EFT evolution to salvage information, maintaining the unyielding boundary drawn by the small corrections theorem.

C. A Non-Conventional Loophole: Superposed Vacua at Early Times

The discussion so far has treated the late Page-time regime as the place where $U_{t,\rm EFT}(\Delta t)$ might depart from $U_{\rm EFT}$ by an $\mathcal{O}(1)$ amount. A different kind of objection can be raised about early times instead, and it is worth addressing directly, since it does not resemble any of the resolutions considered elsewhere in this paper.

Write the early black hole state in the same branch decomposition used above:

$$|\Psi_{\text{early BH}}\rangle = \alpha|\psi_{\text{EFT-like}}\rangle + \beta|\psi_{\text{QG-like}}\rangle. \tag{6}$$

So far we have assumed that whatever $|\psi_{\text{QG-like}}\rangle$ denotes at early times, $U_{t,\text{EFT}}(\Delta t)$ restricted to that branch is itself only an ϵ-departure from U_{EFT}, consistent with the theorem's premise. But this need not be the case. A collapsing star can, in principle, be written as a superposition in which one branch (with $|\alpha|^2 \approx 1$) collapses to form a black hole in the ordinary way, while another, heavily suppressed branch corresponds to the same matter simply failing to form a horizon at all, dispersing instead into what is, for practical purposes, ordinary Minkowski spacetime. One can picture this as the amplitude for the would-be Schwarzschild radius to be effectively zero. In that branch, $|\psi_{\text{QG-like}}\rangle$ is not a small correction to the black-hole vacuum; it is a different vacuum sector entirely, and $U_{t,\text{EFT}}(\Delta t)$ restricted to it can differ from the black-hole generator U_{EFT} by an $\mathcal{O}(1)$ amount even at $t = 0$.

This looks, at first glance, like a violation of the theorem's premise that early corrections stay small. It is not, for two independent reasons.

First, $|\psi_{\text{EFT-like}}\rangle$ and $|\psi_{\text{QG-like}}\rangle$ here are not just approximately but macroscopically distinct configurations, so $\langle\psi_{\text{EFT-like}}|\psi_{\text{QG-like}}\rangle \approx 0$ to an excellent approximation, for the same reason that a macroscopic collapsing star does not appreciably tunnel into a completely different classical configuration. The unitarity-plus-orthogonality argument given earlier in this section applies immediately: α and β cannot be reassigned by the dynamics without violating this near-orthogonality, so $|\beta|^2$ stays astronomically small throughout, at least in this simplified two-branch picture. The entanglement entropy actually tracked by the small corrections theorem is the entropy of the dominant, observed branch, and that branch's evolution is governed by U_{EFT} to within ϵ, exactly as the theorem requires. The suppressed branch's radically different internal dynamics never enters the calculation at any appreciable weight.

A more fine-grained picture makes an eventual $\mathcal{O}(1)$ change in $|\alpha|^2$ considerably less mysterious, once it is understood at the level of individual microstates rather than as one coherent shift of the whole branch. Write $|\psi_{\text{EFT-like}}\rangle$ itself as a superposition over the many black hole microstates k compatible with the same coarse macroscopic data:

$$|\psi_{\text{EFT-like}}\rangle = \sum_k \alpha_k |\psi_{k,\text{EFT-like}}\rangle. \tag{7}$$

This is exactly the point at which ETH does its work: to an outside observer restricted to simple observables, this entire superposition is indistinguishable from a single ordinary EFT state, regardless of how the weight is distributed across k. For most k,

$$U_{t,\text{EFT}}(\Delta t)|\psi_{k,\text{EFT-like}}\rangle \approx U_{\text{EFT}}|\psi_{k,\text{EFT-like}}\rangle, \tag{8}$$

consistent with everything argued above. But it is conceivable that for some microstates the true dynamics instead produces a complete transition,

$$U_{t,\text{EFT}}(\Delta t)|\psi_{k,\text{EFT-like}}\rangle = |\psi'_{k,\text{QG-like}}\rangle, \tag{9}$$

rather than a small correction to $|\psi_{k,\text{EFT-like}}\rangle$ itself. When this happens, that microstate's entire weight α_k is transferred to the QG-like sector, not diluted into an ϵ-sized correction. Summed over enough such microstates, this can build an appreciable shift in the aggregate

$$|\alpha|^2 = \sum_k |\alpha_k|^2, \tag{10}$$

even though ETH keeps the coarse-grained EFT description looking essentially unchanged to simple observables until a genuinely significant fraction of the microstates has transitioned. This transition effect does not weaken the paper's central requirement that a macroscopic departure from EFT be genuinely dynamical rather than merely kinematic, since each such transition is itself a fully $\mathcal{O}(1)$ dynamical event at the level of that microstate, exactly the kind of large, per-branch departure from U_{EFT} that the small corrections theorem and the Page Curve Norm Requirement already demand. What it supplies is a concrete, microstate-resolved picture of how such a departure could accumulate gradually at the coarse-grained level while remaining discontinuous, branch by branch, underneath.

Second, nothing about the suppressed branch itself constitutes a breakdown of EFT logic. $U_{t,\text{EFT}}(\Delta t)$ restricted to $|\psi_{\text{QG-like}}\rangle$ is not non-linear, non-local, or non-unitary; it is simply the generator appropriate to an empty, horizon-free spacetime, an entirely ordinary local EFT in its own right. There is no drama in either branch individually. An empty-spacetime branch trivially has no lasting radiation entanglement to account for, since no evaporation is occurring in it at all. What has happened is not that the effective description has failed, but that the full state contains two macroscopically distinct, individually well-behaved EFT sectors, one of which overwhelmingly dominates.

Conventional presentations of black hole complementarity, the Petz map and entanglement wedge reconstruction, ER=EPR, and replica wormholes do not frame themselves this way; they are formulated as accounts of how information escapes from a black hole that is assumed to exist throughout, correcting the near-horizon dynamics or the entanglement structure of the radiation of that one black hole, rather than as a superposition over horizon formation itself. That said, we do not rule out that a suitably reformulated, microstate-resolved version of one or more of these proposals could turn out to connect to the mechanism just described. New saddle points contributing to the gravitational path integral, of the kind invoked by islands and replica wormholes, are natural candidates for such a reformulation, since a new saddle dominating the path integral at late times is not obviously different, at the level of individual microstates, from the kind of complete $|\psi_{k,\text{EFT-like}}\rangle \to |\psi'_{k,\text{QG-like}}\rangle$ transition described above. We leave this identification for future work. Either way, the scenario considered in this subsection reduces to an early-time instance of the same dichotomy already established above: only if β is driven toward $\mathcal{O}(1)$, the same kind of event catalogued as the Macroscopic Dynamic Phase Transition option, does it threaten the small corrections theorem's conclusion, and that possibility is already accounted for.

IV. EXTENDING THE SMALL CORRECTIONS THEOREM: THE PAGE CURVE NORM REQUIREMENT

The small corrections theorem establishes that if the dynamic perturbation V has small operator norm, the entanglement entropy of the radiation cannot deviate significantly from the EFT prediction for the given near-semiclassical initial state. This section strengthens that result in a complementary direction, independent of the no-drama argument of Section III A. The *Page Curve Norm Requirement* shows that the Page curve's demand for an $\mathcal{O}(1)$ deviation from the monotonically increasing semiclassical entropy forces the operator norm of any unitary perturbation implementing it to be macroscopically large, regardless of what state it acts on.

A. Shared Setup

Let the bipartite Hilbert space be $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$, where $d = \min(d_A, d_B)$ is the effective dimension of the EFT subspace. Let $H_{\text{exact}} = H_{\text{EFT}} + V$, where V is the interaction perturbation acting across the bipartite boundary, with $\mathrm{Tr}_A(V) = 0$ and $\mathrm{Tr}_B(V) = 0$ to exclude trivial local terms. The reduced density matrix and its von Neumann entropy are $\rho_A = \mathrm{Tr}_B(\rho_{AB})$ and $S_A = -\mathrm{Tr}(\rho_A \log \rho_A)$.

Under the unperturbed semiclassical dynamics, H_{EFT} dictates an entropy evolution rate of $dS_A/dt|_{\text{EFT}}$. The full Hamiltonian H_{exact} modifies this rate. Because the time derivative of the von Neumann entropy is linear with respect to the Hamiltonian commutators, the deviation in the entropy generation rate, defined as $\delta \dot{S}_A \equiv dS_A/dt|_{\text{exact}} - dS_A/dt|_{\text{EFT}}$, isolates the contribution of V:

$$\delta \dot{S}_A = i\,\mathrm{Tr}_A \left(\log \rho_A \cdot \mathrm{Tr}_B([V, \rho_{AB}]) \right). \tag{11}$$

To first order in Δt, the correction to the entropy change over one emission step compared to the pure EFT evolution is $\delta S_A \approx \delta \dot{S}_A \Delta t$.

B. Theorem: Page Curve Requires Large Operator Norm

Premise. The Page curve requires that after the Page time, the exact entanglement entropy of the radiation must decrease to eventually return to a pure final state, while the EFT prediction continues to grow by $\mathcal{O}(1)$ per emission step. Therefore, the perturbation must supply an $\mathcal{O}(1)$ negative correction to overcome the semiclassical background. There must exist a physical state $|\psi_{\text{Page}}\rangle$ for which the deviation is:

$$\delta S_A(|\psi_{\text{Page}}\rangle) = -\mathcal{O}(1). \tag{12}$$

Claim. Any unitary perturbation V implementing such a Page curve turnaround must have an operator norm $\|V\| = \mathcal{O}(1/\Delta t)$, i.e., macroscopically large on the emission timescale.

Proof. The deviation in the von Neumann entropy rate satisfies a rigorous upper bound in terms of the operator norm of the generating interaction V. Specifically, using the bound on the commutator $\|[V, \rho_{AB}]\| \leq 2\|V\|$ and the contractivity of the trace norm, one obtains:

$$\left| \delta \dot{S}_A \right| \leq K(d) \cdot \|V\|, \tag{13}$$

where $K(d)$ is a constant at most logarithmic in d, of the kind established for bipartite entangling rates [7, 8]. The premise demands that the physical state achieves a correction of $|\delta S_A| = \mathcal{O}(1)$ over one emission step Δt, so the anomalous rate satisfies $|\delta \dot{S}_A| \geq \mathcal{O}(1)/\Delta t$. Combining this with the upper bound yields:

$$\|V\| \geq \frac{1}{K(d)} \cdot \frac{\mathcal{O}(1)}{\Delta t}. \tag{14}$$

Q.E.D. Physical interpretation. Since $K(d)$ grows at most logarithmically in d, this bound is $\mathcal{O}(1)$ on the Δt timescale up to that same slowly growing factor: macroscopically large in absolute terms, whatever the dimension of the EFT subspace. This result requires only that *one* state (the post-Page black hole) undergoes an $\mathcal{O}(1)$ deviation from the EFT entropy trajectory, and it immediately forces a macroscopic operator norm. No fine-tuning of the initial state, no appeal to ETH, and no special microstate structure can circumvent this: the norm bound is a property of the operator V itself, not of the state it acts on. Any proposed resolution that claims to produce the Page curve turnaround while maintaining approximately semiclassical dynamics, meaning keeping $\|V\|$ small, is therefore mathematically ruled out by this theorem independently of the small corrections theorem, and independently of the no-drama argument of Section III A.

V. EVALUATING PROPOSED RESOLUTIONS THROUGH THE DYNAMIC LENS

The small corrections theorem, together with the no-drama response to ETH and the Page Curve Norm Requirement, forces a stark choice: either accept information loss, or accept a macroscopic breakdown of local effective field theory dynamics at the horizon. It is instructive to evaluate various proposed resolutions to the firewall paradox [9] through this lens of kinematic versus dynamic corrections. As we will see, the most prominent physical resolutions implicitly concede this argument's core by introducing precisely the kind of $\mathcal{O}(1)$ dynamic alterations to semiclassical evolution that it demands.

A. Black Hole Complementarity and the Petz Map

Black hole complementarity (BHC), proposed by Susskind, Thorlacius, and 't Hooft [10], offers a framework that attempts to preserve both unitarity and the equivalence principle.

BHC postulates that information is both reflected at the horizon (for the distant observer) and passes smoothly through the horizon (for the infalling observer), avoiding the no-cloning theorem simply because no single observer can access both descriptions simultaneously.

However, the small corrections theorem fundamentally challenges this peaceful coexistence. If we apply the axiom of BHC that local, semi-classical EFT correctly describes the dynamics outside the stretched horizon, then the evolution operator U_{EFT} dictates a continuous monotonic increase in entanglement. Thus, without introducing a macroscopic $\mathcal{O}(1)$ dynamic breakdown to the local Hamiltonian, unitarity cannot be satisfied. BHC attempts to bypass the contradiction by declaring the states kinematically complementary, but the small corrections theorem demonstrates that the *dynamic evolution* required by the two descriptions is mathematically incompatible. The no-drama argument of Section III A further sharpens this: the claim that EFT dynamics hold, no-drama, outside the stretched horizon is precisely the claim that some unitary EFT with small δU governs that region, from which it follows that entanglement entropy cannot decrease there, regardless of how complex the underlying microstate is.

Recently, critics have attempted to revive and rigorously formalize BHC through the lens of quantum error correction (QEC). This program builds on the tensor-network toy models of Pastawski, Yoshida, Harlow, and Preskill [11], which first cast bulk reconstruction as an isometry from a bulk code subspace onto the boundary Hilbert space, and was developed further by Almheiri, Dong, and Harlow [12] and, through the Petz recovery map specifically, by Cotler et al. [13]. (Recent proposals include Cao, Cheng, Karthikeyan, Li, and Preskill [14], along with its analysis by Witten [15]; these recent proposals do not change the analysis below.) In this modern view, the black hole interior is treated as a logical subspace encoded within the physical degrees of freedom of the exterior radiation and the remaining black hole. Even if local unitarity (or final purity) appears to break down in the full Hilbert space, restricting the code space strictly to the exterior subspace allows the Petz map to reconstruct interior operators perfectly. Critics argue that this mathematical reconstruction vindicates BHC: unitarity is restored from the perspective of the exterior observer without needing to explicitly modify the semiclassical dynamics of the infalling observer.

Viewed through the kinematic versus dynamic distinction, the Petz map defense does not evade these results, though the reason is simpler than any detailed mechanism. The Petz map itself is a purely kinematic recovery operation, a quantum channel applied to an already-given, highly entangled state; on its own it says nothing about the operator that actually generates that state as the black hole evaporates. For the reconstruction it describes to be more than a formal bookkeeping device, the true generator $U_{t,\mathrm{EFT}}(\Delta t)$ must, at some point in the evaporation, implement a non-local mapping between interior and exterior degrees of freedom, exactly the kind of $\mathcal{O}(1)$ departure from U_{EFT} that the small corrections theorem and the Page Curve Norm Requirement already require. Invoking the Petz map does not remove this requirement; it is one language for describing what such a departure looks like from the reconstructing observer's point of view. The microstate-level mechanism of Section III C offers one candidate picture of how this could be realized dynamically, with individual microstates undergoing complete transitions rather than the whole state receiving a uniform small correction, though we do not claim that identification is established. Section III B covers separately why $U_{t,\mathrm{EFT}}(\Delta t) \approx U_{\mathrm{EFT}}$ at early times together with $U_{t,\mathrm{EFT}}(\Delta t) \neq U_{\mathrm{EFT}}$ at late times does not evade the small corrections theorem.

A third possibility, already catalogued in Section III B, is that these code constructions are really signaling a breakdown of the bulk effective Hilbert space factorization itself, rather than either a microstate-level transition or genuine bulk backreaction. In that case the Page curve, as a bulk quantity, may not really exist, and the entropy turnaround these codes reproduce is better read, per Interpretation B of that section, as a statement about the boundary CFT rather than a strictly local bulk reconstruction, or, per Interpretation A, as non-factorizability inherited from the boundary from the start. Either reading is still a concession to the theorem.

B. Non-locality: ER=EPR and Islands

Another major class of resolutions attacks the strict tensor product factorization of the Hilbert space. The ER=EPR conjecture [16] of Maldacena and Susskind proposes that quantum entanglement (Einstein-Podolsky-Rosen) and geometrical wormholes (Einstein-Rosen bridges) are fundamentally identical. In this framework, the early Hawking radiation is geometrically connected to the black hole interior. Therefore, when a new Hawking quantum is emitted, its interior partner is not strictly local to the horizon, but is non-locally connected to the distant early radiation.

Recent advancements have formalized this non-local geometry via the calculation of the

Page curve using replica wormholes and the "island" formula [17]. In this formulation, the gravitational path integral includes topologically connected wormhole saddles that dominate after the Page time. The consequence is that the entanglement wedge of the outgoing radiation includes an "island" situated deep inside the black hole interior.

Both ER=EPR and the islands proposal operate on the same philosophical premise: information escapes because the degrees of freedom inside and outside the black hole do not dynamically evolve independently. The local, semiclassical Hamiltonian U_{EFT} assumes that spacelike-separated regions (the interior and the distant radiation) commute and evolve autonomously, precisely the condition that validates the tensor product structure assumed throughout this paper. By introducing wormholes and islands, these models declare that the true QG evolution operator U_{exact} involves macroscopic, non-local dynamic interactions, i.e., $\|V\| \sim 1$. The small corrections theorem is thus vindicated: local EFT dynamics must break down to save unitarity.

C. Operational Limitations and Computational Complexity

Some approaches seek to resolve the paradox not by altering fundamental physics, but by questioning the operational realizability of the paradox itself. Harlow and Hayden argued [18] that for an infalling observer to verify the entanglement between the early radiation and the late Hawking quantum, they would need to perform a quantum computation that takes longer than the black hole's evaporation time, with a decoding time exponential in the entropy of the black hole. Similarly, it has been shown that an infalling observer's causal patch cannot encompass the entire interior of the black hole, making it impossible to collect the information required to verify the interior entanglement.

While these arguments provide an epistemic shield against the firewall (preventing any single observer from ever experiencing the contradiction), they do not address the ontological question of the state's dynamic evolution. Even if the calculation is practically impossible, the objective physical evolution of the state must still be described, and the operator norm $\|V\|$ of the true Hamiltonian's interaction term is an objective physical quantity, irrespective of computational accessibility. Thus, computational complexity arguments bypass the kinematic/dynamic distinction rather than resolving it.

VI. CONCLUSION

The small corrections theorem is not merely a mathematical curiosity within an outdated semiclassical paradigm. The two results developed in this paper extend that diagnostic in complementary directions. The no-drama argument closes the ETH loophole by showing that microstate complexity alone cannot stop an old black hole's horizon from being governed by some unitary EFT generator, so the entropy-growth logic keeps applying regardless of how the underlying state is dressed. The Page Curve Norm Requirement independently shows that producing the Page curve requires a macroscopic perturbation operator norm, without appeal to any particular state. Together, these results dismantle the claim that quantum gravitational state complexity alone resolves the paradox: ETH may explain why microstates look alike to a low-energy observer, but it cannot explain how information escapes an approximately semiclassical Hamiltonian, and the norm requirement shows separately that the escape cannot be done with small perturbation $\|V\|$. Whatever one makes of this diagnostic, it isolates the real cost of solving the information problem: the dynamic evolution at the horizon cannot remain an ϵ-perturbation of EFT, and the horizon must at some point genuinely stop being no-drama. Whether through gravitational shockwaves, non-local replica wormholes, fuzzball horizon replacements, or the breakdown of Hilbert space factorization via subregion holography, the dynamics themselves must be radically altered. The small corrections theorem demands a fundamental rewriting of local spacetime physics at the horizon, and the no-drama argument and the Page Curve Norm Requirement give that demand its precise, state-independent form.

DATA AVAILABILITY AND DECLARATION OF INTERESTS

The author(s) have no funding source to declare. Furthermore, there is no conflict of interests.

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